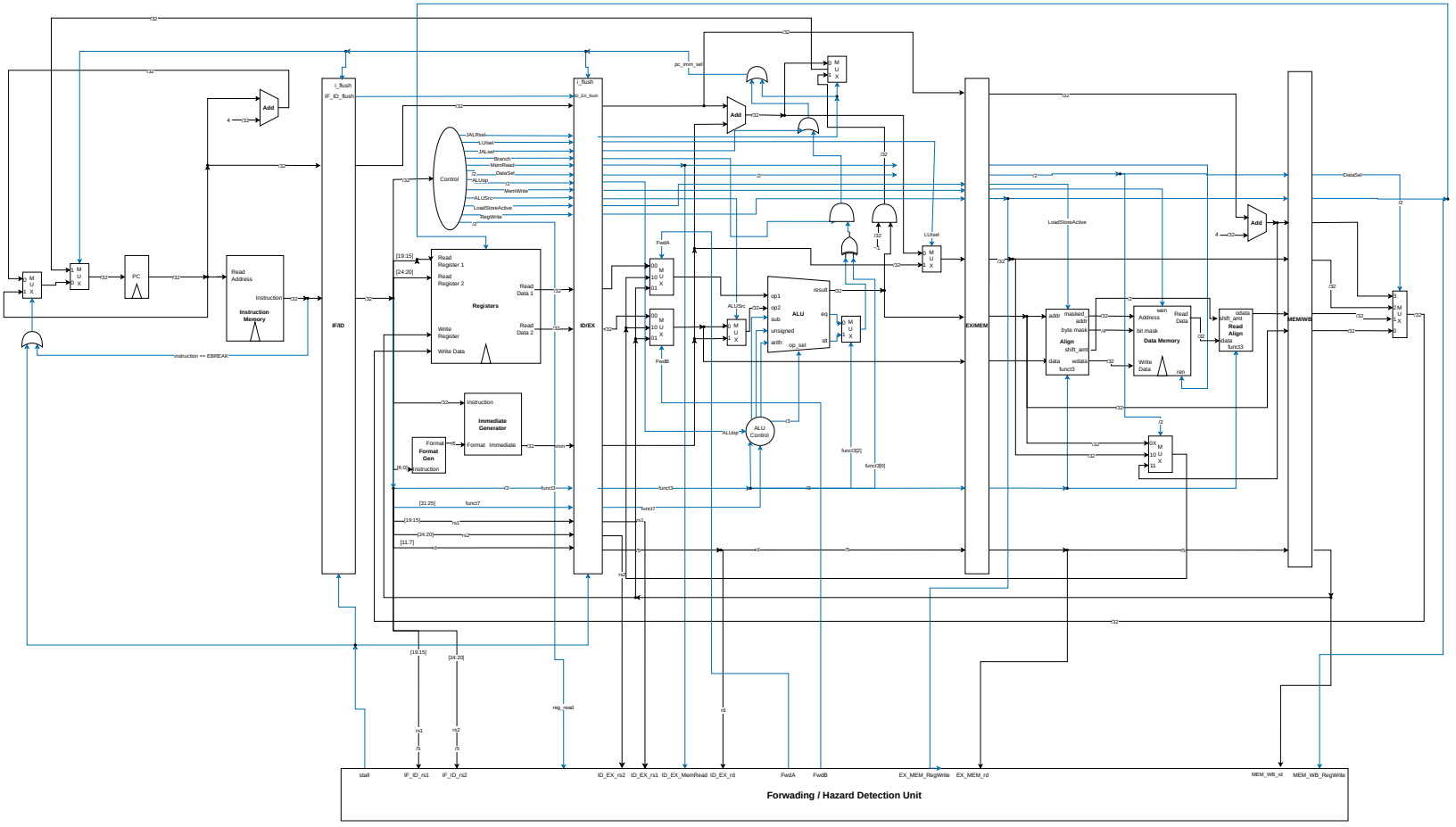

Add
PC
Instruction Memory
Read Address
Instruction
IF/ID
Control
Registers
Read Register 1
Read Register 2
Write Register
Write Data
Read Data 1
Read Data 2
Immediate Generator
Format Gen Instruction
ID/EX
ALU
ALU Control
EX/MEM
Data Memory
Align
Read Align
MEM/WB
Forwading / Hazard Detection Unit
stall
IF_ID_rs1
IF_ID_rs2
ID_EX_rs2
ID_EX_rs1
ID_EX_MemRead
ID_EX_rd
FwdA
FwdB
EX_MEM_RegWrite
EX_MEM_rd
MEM_WB_rd
MEM_WB_RegWrite